Geophysics IIIA

Practical: Gravity Methods

May 27, 2026

Instructions

This problem set explores the physical origin and interpretation of gravity anomalies and their relationship to crustal structure and isostasy. You may use forward modeling software (Talwani solution) and may write short scripts to automate model construction and parameter testing. You are encouraged to use AI tools to assist with coding, provided you understand and explain the models you produce.

Question 1: Gravity Anomalies of Simple Bodies [10 pts.]

This question builds on **Week 4, Activity 2: Structure Density Reasoning**.

(A) Conceptual predictions

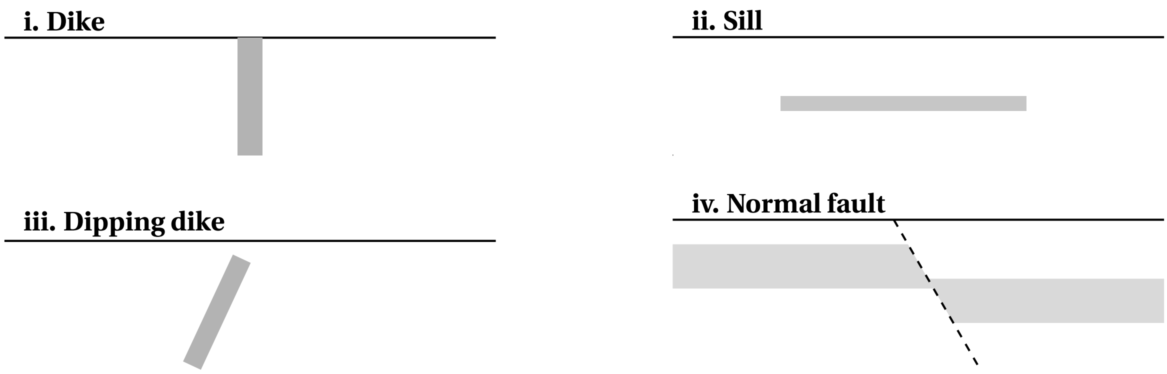


Figure 1: Idealized 2-D density structures used to explore gravity anomaly geometry.

Sketch the expected gravity anomaly as a function of horizontal distance. On each sketch, clearly indicate:

1. The location of the maximum anomaly relative to the body
2. Whether the anomaly is symmetric or asymmetric
3. The far-field (asymptotic) behavior

(B) Forward modeling

Using the Talwani gravity modeling program (`gravity2d_gui.py`) to:

1. Draw polygons to match the the bodies above.
2. Compute the gravity anomaly assuming a constant density contrast.
3. Compare the modeled anomalies with your sketches.
4. Explore changes in the dike model. Comment specifically on how the following changes the wavelength and amplitude:
    (a) depth of burial

(b) density contrast
(c) width of the body

Question 2: Shallow and Deep Bodies [10 pts.]

In this question you will compute gravity anomalies using the analytic solution for a two-dimensional body of constant density contrast, following the formulation of Talwani et al. (1959). All bodies are assumed to extend infinitely perpendicular to the x–z plane.

(A) Gravity anomaly of a 2-D polygon

For a polygon with N vertices (x_i, z_i) ordered clockwise, the vertical gravity anomaly at surface point x due to a body of density contrast $\Delta\rho$ is

$$\Delta g(x) = 2G\Delta\rho \sum_{i=1}^{N} \left[(x_i - x) \ln\left(\frac{r_{i+1}}{r_i}\right) + (z_{i+1} - z_i) \left(\tan^{-1}\frac{x_{i+1} - x}{z_{i+1}} - \tan^{-1}\frac{x_i - x}{z_i} \right) \right],$$

where

$$r_i = \sqrt{(x_i - x)^2 + z_i^2}.$$

You may assume $G = 6.67 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$. This expression is exact for any 2-D polygonal body.

This problem will be easiest to execute in python or matlab.

(B) Triangular rift basin

Consider a triangular shaped basin with two opposite 60° dipping fault planes, filled to the top with sediments.

1. Compute the coordinates for a vertical offset of 5 km. Place (0,0) in the center of the basin.
2. Use the equation above to compute the gravity anomaly.
3. Plot $\Delta g(x)$ and identify:
   - the location of the maximum gradient,
   - the direction of asymmetry.

(C) Trapezoidal rift basin

The trapezoid should have sloping edges comparable in dip to in part (B). Make the depth of the basin 3 km in this case with the same surface coordinates for the basin.

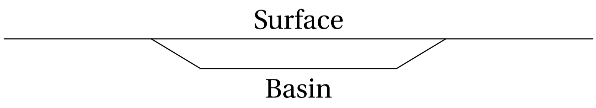
1. Compute the gravity anomaly using the polygon formula.
2. Compare the trapezoidal and triangular basin gravity models.
3. Discuss which features of the anomalies are controlled by:
   - the slope of the edges,
   - the depth extent of the body,
   - the density contrast.

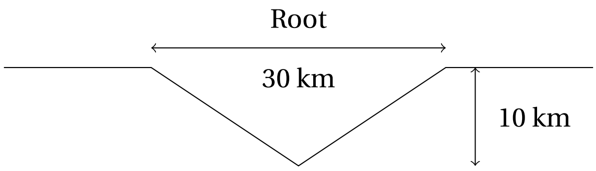
(D) Basin over a crustal root

Model a shallow sedimentary basin overlying a deeper crustal root. Represent the crustal root as a triangular body with:

- Maximum thickness: 10 km
- Total width: 30 km

1. Compute the gravity anomaly of the root alone.
2. Compute the anomaly of a shallow basin alone.





3. Combine the basin and root in a single model.

By adjusting basin parameters (depth, width, density contrast), determine whether a basin-only model can replicate a root-only model.

Explain what this implies about non-uniqueness in gravity interpretation.

Question 3: Isostasy — Oceanic vs Continental Lithosphere [10 pts.]

Complete the **Ocean vs. Continent Isostasy** problem from Week 5.

You will show:

1. why crust alone is insufficient to balance the columns,
2. determine the difference in density contrast between oceanic and continental lithoshere to account for the difference in elevation, and
3. determine the compositional change required to balance the columns.

Question 4: Isostatic Response to Erosion [10 pts.]

Consider a hypothetical mountain belt initially in local isostatic equilibrium. Unless otherwise stated, assume Airy isostasy and neglect flexural support.

Include all assumptions and justify your parameter choices.

(A) Single-layer crust (static response)

Assume a single crustal layer of uniform density over mantle. The initial surface elevation of the mountain belt is $\varepsilon_0 = 6100$ m, and the total crustal thickness is approximately ~ 70 km.

1. If 2 km of material is eroded instantaneously from the surface, what is the new equilibrium surface elevation after isostatic rebound?
2. How much total erosion is required to return the surface elevation to the surrounding baseline?
3. What fraction of eroded material is replaced by isostatic rebound?

Clearly state the densities you assume for the crust and mantle, and show all steps.

(B) Continuous erosion with isostatic rebound

Now consider *continuous erosion* acting simultaneously with isostatic rebound. Assume that the erosion rate is proportional to surface elevation,

$$\frac{dE}{dt} = k\varepsilon(t),$$

where $E(t)$ is the cumulative erosion thickness, $\varepsilon(t)$ is the surface elevation, and k is an erosion constant with units of inverse time.

Reasonable values for k in continental mountain belts are $k \sim 0.05$–0.5 Myr^{-1}. For this problem, use

$$k = 0.1 \text{ Myr}^{-1}.$$

Under local isostatic equilibrium, the rate of change of surface elevation due to erosion and rebound may be written in the general form

$$\frac{d\varepsilon}{dt} = -\alpha\frac{dE}{dt},$$

where α depends on the isostatic model and density structure.

1. Derive the value of α for Airy isostasy in terms of crustal and mantle densities.
2. Combine the equations above to obtain a differential equation for $\varepsilon(t)$.

The resulting equation has the general solution

$$\varepsilon(t) = h_0 \exp(-\alpha k t),$$

where h_0 is the initial elevation at $t = 0$.

3. Show that in the absence of a density contrast (no rebound) the solution reduces to $\varepsilon(t) = \varepsilon_0 \exp(-kt)$
4. What is the change in the initial erosion rate with and without isostasy?
5. Using your derived value of α, calculate the time required to erode 95% of the initial relief (i.e. for $\varepsilon = 0.05\,\varepsilon_0$).

6. Compare this timescale to the erosion-only timescale k^{-1}, and explain physically why they differ.
7. Briefly discuss what this simple model implies about the longevity of large mountain belts.